

P4 — Design-Aware Main Estimate (Lab 5)

Model submission — demonstration paper (Kueh et al., BMJ Open 2026).

Estimand convention, used identically in every increment.

- **Scientific (locked) estimand — IV vs III.** Non-obese with high central adiposity against non-obese with low central adiposity. This isolates central adiposity with body-mass category held fixed, and it is the question the term is built on.
- **Replication estimand — IV vs I.** The paper’s own parameterization, kept because reproducing it is what *Replicate* means.

Both are contrasts within one fitted model. Every estimate below states which it is.

The estimand, stated before the model

We fix the target of inference first, because “the association” is under-specified until we name a population. Our contrast is the all-cause mortality hazard of phenotype **Group IV** (non-obese by BMI, high central adiposity WHtR $\geq$ 0.6) relative to **Group I** (obese, low central adiposity), among adults who meet FLI-based MASLD criteria, over follow-up censored 31 December 2019 (median 6.7 years). The measure is a hazard ratio.

The design question is *which* population that HR describes. The paper’s unweighted Cox model estimates a **sample-conditional** association inside the 6,371 people who happened to be sampled. But NHANES is a stratified, clustered, unequal-probability survey; without weights the estimate is not a statement about US adults with MASLD. Making the estimand *population-average* — the target-of-inference we think the paper actually wants — requires the survey machinery. So P4 is not “add weights and see if the number moves.” It is a change of estimand, and we report both because both are legitimate answers to *different* questions.

Replicate: the unweighted Cox model

We first reproduce the paper’s Table 2 on the full cohort (n=6,371; 586 all-cause deaths), Group I as reference. Group IV is the headline “worst survival” phenotype:

Model	Group II	Group III	Group IV
Unadjusted	6.68 (2.76–16.14)	6.23 (2.52–15.41)	15.14 (6.19–37.04)
Model 1 (clinical)	2.50 (1.03–6.09)	3.23 (1.30–8.01)	3.28 (1.33–8.11)
Model 2 (+ smoking, sedentary)	2.37 (0.97–5.76)	2.99 (1.20–7.43)	2.91 (1.18–7.20)

These match the publication to rounding (paper IV: 15.13 unadjusted, 2.89 in Model 2). The replication is clean: the fifteen-fold crude hazard collapses to roughly three-fold once age and cardiometabolic burden are adjusted — most of the crude signal is confounding by who lands in each phenotype — but Group IV remains the worst group and its Model-2 confidence interval still excludes 1.

Improve: a design-aware estimate

The analytic cohort is the NHANES **fasting subsample**, because the FLI needs fasting triglycerides. The correct weight is therefore the pooled fasting-subsample weight, WTSF2YR divided by 6 (six two-year cycles), with SDMVSTRA strata and SDMVPSU clusters. Valid weights exist for **n=6,048 across 184 clusters**. On that same subset the unweighted IV HR is 16.96 (slightly higher than the full-cohort 15.14 — the weight-eligible subset is not identical to the full cohort, which itself matters).

Survey-weighted, all-cause, Group I reference:

Group	Weighted HR (95% CI)
II	5.38 (1.19–24.39)
III	4.04 (0.89–18.34)
IV	12.40 (2.69–57.15)

Because Group I is a tiny, low-event reference (n=391, 5 deaths), the vs-I contrasts are unstable in either weighting scheme. So we also estimate **weighted direct contrasts** against the other high-risk groups, which is the comparison a clinician actually cares about:

- **IV vs III: 3.07 (2.11–4.46)** — non-obese/high-central vs non-obese/low-central
- **IV vs II: 2.30 (1.76–3.02)** — non-obese/high-central vs obese/low-central

Those three are **crude** survey-weighted estimates. The adjusted versions, and the comparison that separates adjustment from design, are in the matrix below.

The matrix that separates adjustment from design

The running ledger of 15.1 → 2.9 → 12.4 mixes crude and adjusted models on different row sets, so it is not a sequence of successive corrections and should not be read as one. To separate **adjustment** from **survey design** the rows must be held fixed.

All four cells below use the **same 5,939 Model-1 complete-case rows** inside the locked domain.

	IV vs I (<i>replication</i>)	IV vs III (<i>locked</i>)
unweighted, crude	16.05 (5.91–43.58)	2.34 (1.72–3.18)
unweighted, Model 1	3.79 (1.38–10.39)	1.06 (0.77–1.45)
weighted, crude	11.93 (2.60–54.84)	3.07 (2.11–4.48)
weighted, Model 1	2.97 (0.69–12.79)	1.29 (0.90–1.86)

Read down a column for what **adjustment** does; read across a row for what **design** does.

- **On the locked estimand, adjustment dominates design.** Adjustment moves 2.34 to 1.06 unweighted and 3.07 to 1.29 weighted. Design moves 2.34 to 3.07 crude and 1.06 to 1.29 adjusted — in the *opposite* direction, and by less.
- **Design changes no conclusion on the locked estimand.** Both adjusted intervals include 1 (0.77–1.45 and 0.90–1.86).
- **Design does change the conclusion on the replication estimand.** Unweighted Model 1 is 3.79 (1.38–10.39), excluding 1; weighted Model 1 is 2.97 (0.69–12.79), which does not.
- So “survey design matters” is true **of the paper’s own contrast**, and much weaker for the question we locked. Anyone ranking design as the dominant threat has to say which estimand they mean.

Interpretation

Two things happen when we move from sample-conditional to population-average. First, the point estimate **attenuates** (IV 16.96 → 12.40): weighting changes how much each respondent contributes according to the sampling design, so representative estimation pulls the number down. Second, the confidence intervals **widen** (IV upper bound 37 → 57). That widening is the **widening of the survey-aware interval** — clustering within 184 PSUs and unequal weights reduce the *effective* sample size below the nominal 6,048, so the honest CI is wider than the unweighted one pretends. The unweighted model was over-confident, not more precise.

What the matrix actually supports. For the **locked IV-vs-III** estimand, adjustment is consequential and survey weighting changes the point estimate modestly (1.06 → 1.29) *without* altering the inferential conclusion — both adjusted intervals include 1, so there is **no clear evidence of a higher hazard at the**

available precision. For the **replication IV-vs-I** estimand, weighting materially changes the uncertainty and whether the interval excludes 1 (3.79, 1.38–10.39 becomes 2.97, 0.69–12.79). Group IV having the largest point estimate describes an *ordering*; it does not establish that central adiposity carries excess mortality among non-obese adults after adjustment.

Which weight: a rule, not a judgment call

An earlier draft of this submission called the weight choice “a genuine puzzle we cannot fully resolve” and argued the MEC weight was equally defensible. **That was wrong, and we withdraw it.**

CDC’s guidance is the **least common denominator rule**: when an analysis draws on variables measured in different components, use the weight belonging to the **smallest** applicable subsample. FLI requires fasting triglycerides, so **WTSAF2YR** governs — even though GGT and anthropometry are available MEC-wide. The rule exists precisely because analyses routinely straddle frames; it is not a tie to be broken by argument.

The MEC weight is not a defensible alternative here: it would weight to a population in which the exposure cannot be constructed at all. There is no sensitivity axis to carry into M2 on this point.